# Vedansh Chaturvedi

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 8.56 |
| T.A.V. Higher Secondary School | Class XII (MPBSE) | 2019 | 93.00% |
| Govt. Model School, Chanderi | Class X (MPBSE) | 2017 | 94.67% |

## WORK EXPERIENCE

**Honeywell Aerospace, Hyderabad** | Software Engineering Intern

- Implemented **AI agents** to streamline production workflows: an autonomous codebase documentation agent and an query agent integrated with internal KMS service.
- Built and orchestrated multi-agent pipeline using Java, **LangGraph**, **Deep Agents**, and **GitHub Copilot SDK.**
- Built an **AI integrated ontology** and **knowledge-graph** platform for KMS service using semantic web technologies (RDF, RDFS, OWL), linking database schemas, code modules, and workflows.

**WiCoN (Wireless Communication and Networking) Lab |** Research Assistant

- Developing an AI-powered hospital digital twin using CV and edge AI to monitor patient flow, room occupancy, and waiting times from CCTV video streams.
- Designing spatio-temporal analytics for bottleneck detection, time-toxicity estimation, and resource utilization analysis.

## PROJECTS

**GraphRAG Explorer**

- Developed **GraphRAG Explorer**, an interactive document question-answering application that visualizes knowledge graphs, provides source-cited answers, replays evidence retrieval, and compares graph-based retrieval with vector search.
- Built a **Java** application, integrating **LangChain4j** for entity extraction and embeddings, **Neo4j GDS** for community detection; implemented Local, Global, DRIFT, and vector retrieval with citation validation and live progress streaming.

**US Accident Severity Prediction and Hotspot Analysis** | PRML (EE5610)

- Built a severity prediction pipeline for **2M+** US traffic accident records, engineering features, then training and comparing **Random Forest**, **XGBoost**, and **LightGBM** classifiers -- achieving **88.1%** accuracy and **0.94** ROC-AUC.
- Conducted geospatial hotspot analysis using heatmaps, and spatial clustering, identifying high-risk cities and weather-related accident patterns to support data-driven traffic safety and infrastructure planning.

**Chest X-Ray Diagnosis & Retrieval**

- Developed a multi-label chest X-ray classifier using **EfficientNet-B2** and **TorchXRayVision**. Integrated **CLIP ViT-B/32** for image-text retrieval. Data analysis and visualization using pandas, seaborn, and matplotlib.
- Achieved **AUROC-macro 0.620** and **macro-F1 0.483** on CheXpert, a heterogeneous clinical dataset, with strong per-label separability on Edema and Cardiomegaly. Applied calibration (LBFGS), test-time augmentation (TTA),

**Stable Diffusion from Scratch**

- Reconstructed the complete **Stable Diffusion v1.5** architecture in PyTorch, implementing the Denoising **UNet**, **Variational Autoencoder (VAE)**, and **CLIP** text encoder from scratch.
- Developed a strong understanding of latent diffusion models, noise scheduling, cross-attention mechanisms.

**Multilingual Emotion Detection**

- implemented a multilingual multi-label emotion classifier using **DistilBERT** and **mBERT** with LSTM+attention and Hugging Face Transformers. F1-macro improved from baseline **0.372 to 0.620** (English) and **0.811** (Multilingual).

- Leverages language-aware weighting, imbalance-aware losses (Focal, Asymmetric Focal, Weighted BCE). Provides interpretability using LIME and Attention visualization.

## TECHNICAL SKILLS

**Languages:** C, C++, Java, Python, SQL

**Frameworks and Libraries**: LangChain, LangGraph, PyTorch, Keras, Scikit-Learn, NumPy, Pandas, Copilot SDK

**GenAI:** AI Agents, AI Harness, MCP, LLMs, RAG, Graph Engineering, Hugging Face Transformers

**Tools and Softwares:** Git, Docker, MATLAB

## ACHIEVEMENTS

**Winner** of Honeywell Aerospace AI Hackathon -- AI-Powered Infrastructure Auto Healer

Achieved LeetCode **Knight** badge **(**Rated **1877)** -- ranked in the **top 6%** globally

Solved **1000+** questions across various platforms including LeetCode, GFG, NeetCode, Codeforces, and CodeChef

## POSITIONS OF RESPONSIBILITY

**Teaching Assistant :** Digital Communications (EE5837) and Automotive Communication and Sensing (EE5640)

## RELEVANT COURSES

Advanced Data Structure and Algorithms, Convex Optimization, Deep Learning, Foundations of Machine Learning, Matrix Theory, Probability and Stochastic Processes, Representation Learning, Stochastic Processes: Theory and Applications

## HOBBIES AND INTERESTS

Formula 1, Psychology, Neuroscience, Badminton, Swimming, Non Fiction Books